# Confidence intervals. Population known, z-score

## Data scientist salary

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| **Background** | You are given the same dataset from the lesson. The population standard deviation is known to be $ 15,000. |
| **Task 1** | Calculate the mean and the standard error |
| **Task 2** | Find the appropriate z-score for calculating a 90% confidence interval |
| **Task 3** | Find the 90% confidence interval |

| Dataset |
|---|
| $ 117,313 |
| $ 104,002 |
| $ 113,038 |
| $ 101,936 |
| $ 84,560 |
| $ 113,136 |
| $ 80,740 |
| $ 100,536 |
| $ 105,052 |
| $ 87,201 |
| $ 91,986 |
| $ 94,868 |
| $ 90,745 |
| $ 102,848 |
| $ 85,927 |
| $ 112,276 |
| $ 108,637 |
| $ 96,818 |
| $ 92,307 |
| $ 114,564 |
| $ 109,714 |
| $ 108,833 |
| $ 115,295 |
| $ 89,279 |
| $ 81,720 |
| $ 89,344 |
| $ 114,426 |
| $ 90,410 |
| $ 95,118 |
| $ 113,382 |

| | | |
|---|---|---|
| **Mean** | $ 100,200 | |
| **Standard error** | $ 2,739 | Population standard deviation/Sqrt count of population number |
| | $ 15,000 | |
| | 5.48 | |
| **90% Confidence interval** | | |

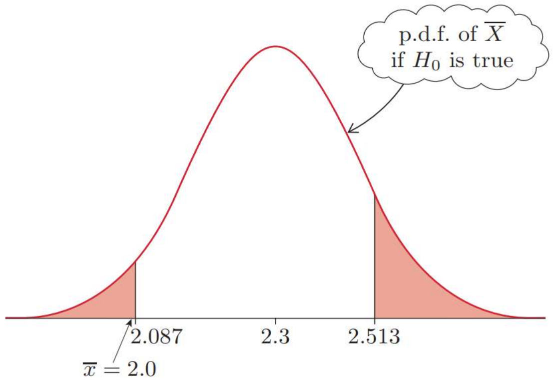
(1-α/2) = (1-0.05) = 0.95

α = 0.1

Look up 0.95 in the table = 1.65 = z

The **standard error** (often shortened to **SE**) is a number that tells us how much we expect a sample average to vary from the true average of the entire population.

$$SE = \frac{s}{\sqrt{n}}$$




**The Significance Level ( alpha):** A percentage that tells us how sure we want to be. Usually, scientists use 5% which is written as α=0.05. It means we want to be 95% sure that our result isn't just a fluke.

**The critical value is Z**. A null hypothesis will be rejected if the test statistic lies on either extreme side of the probability distribution. The middle of the probability distribution is where the values for the null hypithesis are hence the **null hypothesis** is the claim that "everything is normal," and the **critical value** is the exact point where a result becomes "too strange" for the null hypothesis to be true.